

Courier Delivery and Route Tracking System

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Abstract

- The Courier Delivery and Route Tracking System is a Java-based application designed to simplify the management of courier bookings, customers, deliveries, routes, and delivery status.
- Provides a centralized way to record courier, customer, delivery and route details.
- Allows users to add and view couriers and customers and create delivery records.
- Supports searching, tracking current status, updating delivery status and viewing route information.
- Reduces manual record keeping and makes delivery information easier to manage.

Understanding Problem Statement

Problem

Courier, customer and delivery details need to be organized.

Manual record keeping makes delivery information harder to manage.

Users need a simple way to search and track deliveries.

System

Java-based courier management application.

Centralized records for couriers, customers, deliveries and routes.

Menu-driven operations for common tasks.

Main Functions

Add / view / search courier

Add / view customer

Create and track delivery

Update delivery status

View route information

CO 1 Topics – Basic Classes and Data

Courier

Courier ID
Courier Name
Phone Number
Vehicle Number

Customer, Delivery and Route

Customer: ID, name, phone, address
Delivery: ID, customer ID, courier ID, pickup, destination, status
Route: ID, starting point, destination, distance

CO 2 Topics – Main Menu and Operations

Main Menu

1. Add Courier
2. View Couriers
3. Search Courier
4. Add Customer
5. View Customers
6. Create Delivery
7. Track Delivery
8. Update Delivery Status
9. View Route
10. Exit

Delivery Status Flow

New delivery starts with status: Booked

Booked → In Transit

In Transit → Out for Delivery

Out for Delivery → Delivered

Tracking displays courier, destination and current status.

CO 3 Topics – Step-by-Step Development

Development Steps

- Step 1: Add Courier
- Step 2: View Couriers
- Step 3: Search Courier
- Step 4: Add Customer
- Step 5: Create Delivery
- Step 6: Track Delivery
- Step 7: Update Delivery Status
- Step 8: View Route

Testing

- Add 5 couriers and 3 customers.
- Search for a courier.
- Create and track a delivery.
- Change status through the delivery stages.
- Test a non-existing delivery search.
- View route information.

Project Structure and Future Scope

Final Project Structure

CourierDeliverySystem/

- |— Courier.java
- |— Customer.java
- |— Delivery.java
- |— Route.java
- |— Main.java

Conclusion & Future Scope

Provides an organized way to manage courier deliveries.

Helps monitor delivery progress and route information.

Can later be expanded with database storage.

Possible additions: login, live maps, automatic route calculation, notifications and GUI.